

从“方法有效”转向“从理论高度解释为什么有效”，并进行更深入的实验论证。

"Monitoring Mine Hoists" -> "A Novel Boundary-Gated Framework for Extreme Class Imbalance in Non-Stationary Time Series."

1. Formalize the trigger, as a "score > 0", as a constrained optimization problem or a set-theoretic boundary. Define the "State Transition Boundary" mathematically, is it a manifold in the feature space
2. Deepen theory analysis of the "Shift-Aware Graph". Explicitly model the distribution shift between the "Normal" and "Secondary Degradation" states using tools like Optimal Transport or KL-Divergence to justify the trigger mechanism.
3. Theoretical proof or a much deeper analysis of the "Decision Override Logic". Show mathematically how the three constraints (threshold, boundary, precursor) minimize the risk of false positives on common states.

Physics-Informed Machine Learning (PIML)

The “Precursor Constraint” is based on physical logic can be regarded as Physics-Informed Machine Learning (PIML). It is very interesting that deep learning is combined with domain-specific physical constraints. For example, the graph represents state transitions, show how the graph's topology changes as the system degrades.

Experiment

1. Test on other public imbalanced time-series datasets (e.g., UCR archive, NASA's bearing dataset, or Power Grid fault datasets). This proves the model's generalizability beyond mine hoists.
2. The current baselines (TimesNet, iTransformer) are excellent for general time series. For imbalanced learning baselines, compare against cost-sensitive learning, SMOTE-for-time-series, and "Anomaly Detection" models, rare states can be regarded as anomalies.

3. Expand the ablation study. Instead of just removing modules, vary the "strength" of the constraints. Plot a sensitivity analysis of the trigger threshold θ against the Macro-F1 score
4. "Extraction of F2-related Evidence": Use Integrated Gradients or Attention Heatmaps to visualize exactly which sensor signals (e.g., Speed vs. Pressure) trigger the secondary degradation state. This adds the "explainability"
5. Include a section on time/space complexity ($\mathcal{O}$ notation) and inference latency to prove the model is viable for real-time mine safety monitoring
6. Although SGTONet has lower overall accuracy than iTransformer, turn this "weakness" into a "strength" by arguing that in safety-critical systems, the Cost of Misclassification for rare states is infinitely higher than for common states. Define a custom "Economic Loss" or "Safety Risk" metric to prove the model's superiority